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Studierichting: Informatica
Oefeningenreeks 1B nr. 44.2

Opgave: Los de volgende vergelijking op in $\mathbb{C}$

$z^4 - 5(1+2i)z^2 + 24 - 10i = 0$

Stel $g = z^2$. Dit geeft:

$g^2 - (5+10i)g + (24-10i) = 0$

$$\begin{aligned} D &= (-(5+10i))^2 - 4 \cdot 1 \cdot (24-10i) \\ &= -171 + 140i \end{aligned}$$

Resolvente vergelijking opstellen:

$$\begin{cases} x^2 - y^2 = -171 \\ 2xy = 140 \Rightarrow x^2(-y^2) = -(70)^2 = -4900 \end{cases}$$
$\lambda^2 + 171\lambda - 4900 = 0$

$$\begin{aligned} D_\lambda &= 171^2 - 4(1)(-4900) \\ &= 29241 + 19600 = 48841 \end{aligned}$$

$\lambda_1 = \dfrac{-171 + \sqrt{48841}}{2}$ en $\lambda_2 = \dfrac{-171 - \sqrt{48841}}{2} \Rightarrow \lambda_1 = 25$ en $\lambda_2 = -196$

Dus $x^2 = 25$ en $y^2 = 196$

$xy = 70 > 0$, dus:

$g_1 = \dfrac{5 + 10i + (5+14i)}{2}$ *en* $g_2 = \dfrac{5 + 10i - (5+14i)}{2}$

$\Rightarrow g_1 = 5 + 12i$ en $g_2 = -2i$

Voor $g_1 = 5 + 12i$

Resolvente:

$$\begin{cases} x^2 - y^2 = 5 \\ 2xy = 12 \end{cases}$$
$\lambda^2 - 5\lambda - 36 = 0$

$D_\lambda = 25 + 144 = 169 \Rightarrow \sqrt{169} = 13$

$\lambda_1 = \dfrac{5+13}{2}$ en $\lambda_2 = \dfrac{5-13}{2}$

$\Rightarrow \lambda_1 = 9$ en $\lambda_2 = -4$

$x^2 = 9$ en $y^2 = 4$.

Omdat $xy = 6 > 0$: $z_1 = 3 + 2i$ en $z_2 = -3 - 2i$

Voor $g_2 = -2i$

Resolvente:

$$\begin{cases} x^2 - y^2 = 0 \\ 2xy = -2 \end{cases}$$

$\lambda^2 - 0\lambda - 1 = 0$

$x^2 = 1 \Rightarrow x = \pm 1$ en $y^2 = 1 \Rightarrow y = \pm 1$

Omdat $xy = -1 < 0$: $z_3 = 1 - i$ en $z_4 = -1 + i$

oplossingen: $\{3 + 2i, -3 - 2i, 1 - i, -1 + i\}$.

References